

Examining Seasonality in a Human Artifact (Bitcoin) vs. Natural Seasonality (Sunspot Occurrences)

A Comparative Signal Detection Study

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Abstract:

Detecting signals that indicate seasonality is essential in order to recognize patterns in both financial markets and natural phenomena, as it has extensive applications in forecasting and devising strategies. Our goal is to examine the application of signal detection and study of seasonality on two distinct datasets: Bitcoin price fluctuations and sunspot occurrences, exposing the contrasts in noise types and the subsequent impact on analysis. Utilizing the periodogram function in R, noisy datasets can be sifted through via the use of Fourier transform, revealing underlying patterns in the data, which can be used to predict future events or trends.

However, early conclusions revealed that these two datasets have differing noises obscuring signals. While financial market signals are obscured by white noise, signals in nature are often hidden behind sinusoidal noise. Due to this distinction, distinct analytical methods are required in order to obtain accurate signal detection. These findings highlight the necessity of specialized signal detection approaches for various types of noise, with future work centered on refining these methods and exploring additional datasets to further confirm these results, as well as developing new analytical methods for other noise types.

Background Information:

- **Purpose:** To illustrate the significance of studying seasonality in both human-influenced systems, like Bitcoin, and natural systems, like sunspot occurrences, which are unaffected by human activity.
- **Background:** Bitcoin and sunspot values are two fundamentally different datasets. While Bitcoin primarily features human-driven patterns (such as price fluctuations, where noise may result from a wide variety of market forces), natural occurrences exhibit naturally occurring patterns (such as sunspot cycles, which follow natural laws and remain unaffected by human interference).
- Understanding seasonality is crucial in recognizing patterns that guide both human-driven phenomena, such as financial markets, and natural processes, which are unaffected by human influence.
- By exploring what these patterns are caused by and how they can be analyzed, we can garner a better understanding of their seasonality. Signal detection can aid us in forming predictive models, whether it is for financial profit or for early detection of natural events, but only if we recognize what type of noise our data contains, and how that affects our forecasting approach.

Method/Materials:

- **Bitcoin Data:**
 - Imported through R from CryptoCompare, hourly data provided by CCData: open, close, high, low, and volume of Bitcoin
- **Sunspot Data:**
 - Global daily sunspot data since 1818, provided by the Solar Influences Data Analysis Center (SIDC) and accessed through the Sunspot Index and Long-Term Solar Observations (SILSO).
- **RStudio (v.4.4.1)**
 - Graphed Bitcoin price and sunspot occurrence data
 - Matched up timelines for both datasets
 - Packages used: dplyr, lubridate, WaveletComp, TSA, etc.

Analysis Techniques (performed in RStudio):

- **Spectral Analysis:** Periodograms were applied to both datasets to identify dominant frequencies and compare noise characteristics.
- **Wavelet Analysis:** Used to examine time-localized frequency components and reveal any cyclical patterns across both datasets.

Results:

- **Periodogram Analysis:** Display the periodograms for both datasets, focusing on the differences in noise—white noise in Bitcoin prices and sinusoidal noise in sunspot occurrences.
- **Wavelet Analysis:** Present wavelet analysis graphs for both datasets, showing how the nature of the noise influences the periodic signals.
- **Key Findings:** The seasonality in Bitcoin reflects human-influenced volatility, whereas sunspot data shows more consistent, natural cycles.
- Periodogram analysis revealed that Bitcoin prices, driven by market forces, show high-frequency noise patterns, making it difficult to detect long-term seasonality.
- Sunspot occurrences displayed clear cyclical patterns consistent with natural solar activity.
- Wavelet analysis confirmed that the two datasets require different approaches to signal detection, with Bitcoin data obscured by unpredictable white noise and sunspot data presenting well-defined sinusoidal periodicities.

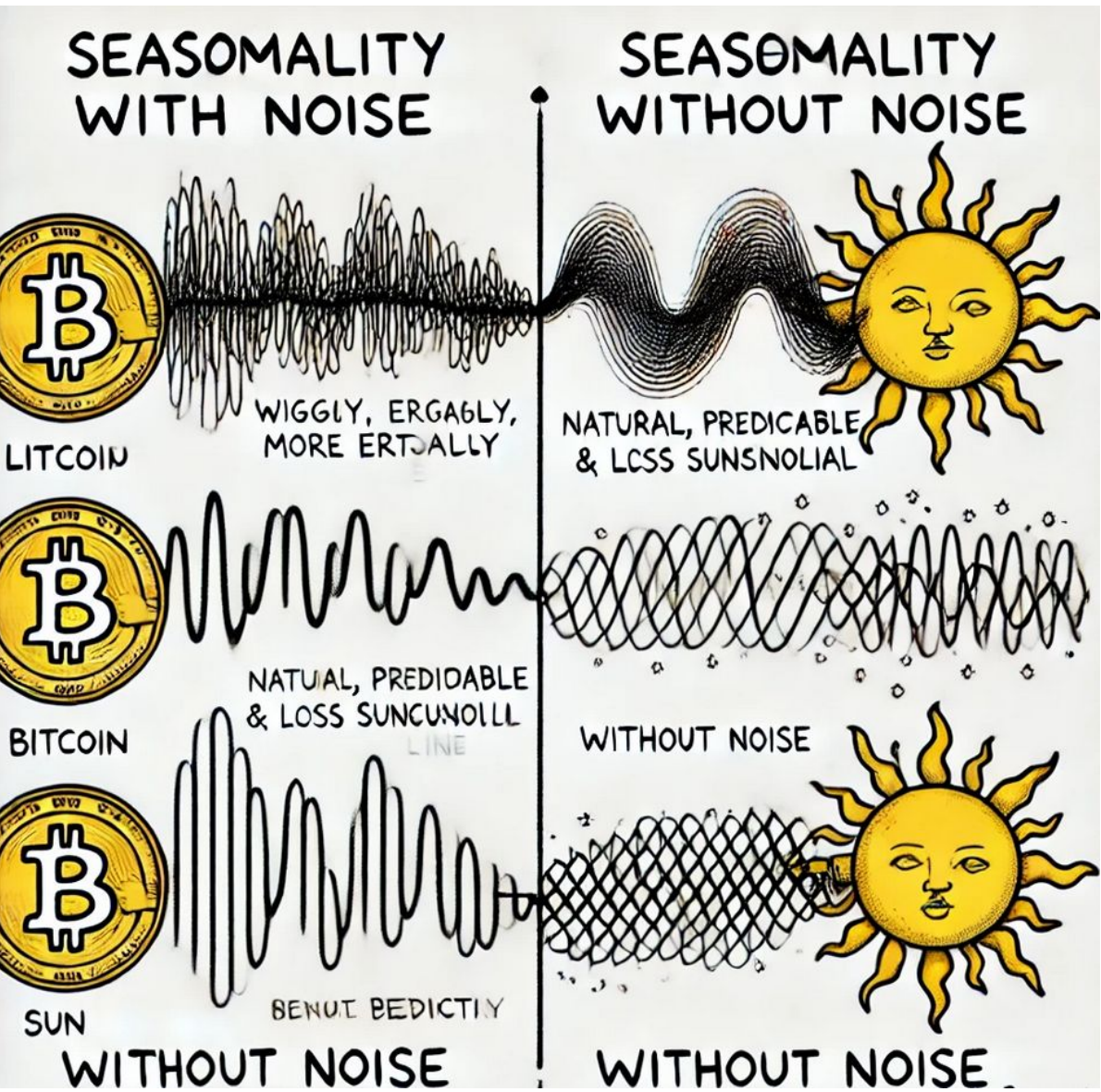


Figure 1: Conceptual figure showing the effects of nse in seasonality of sunspots (unaffected) and bitcoin (affected) by human factors).

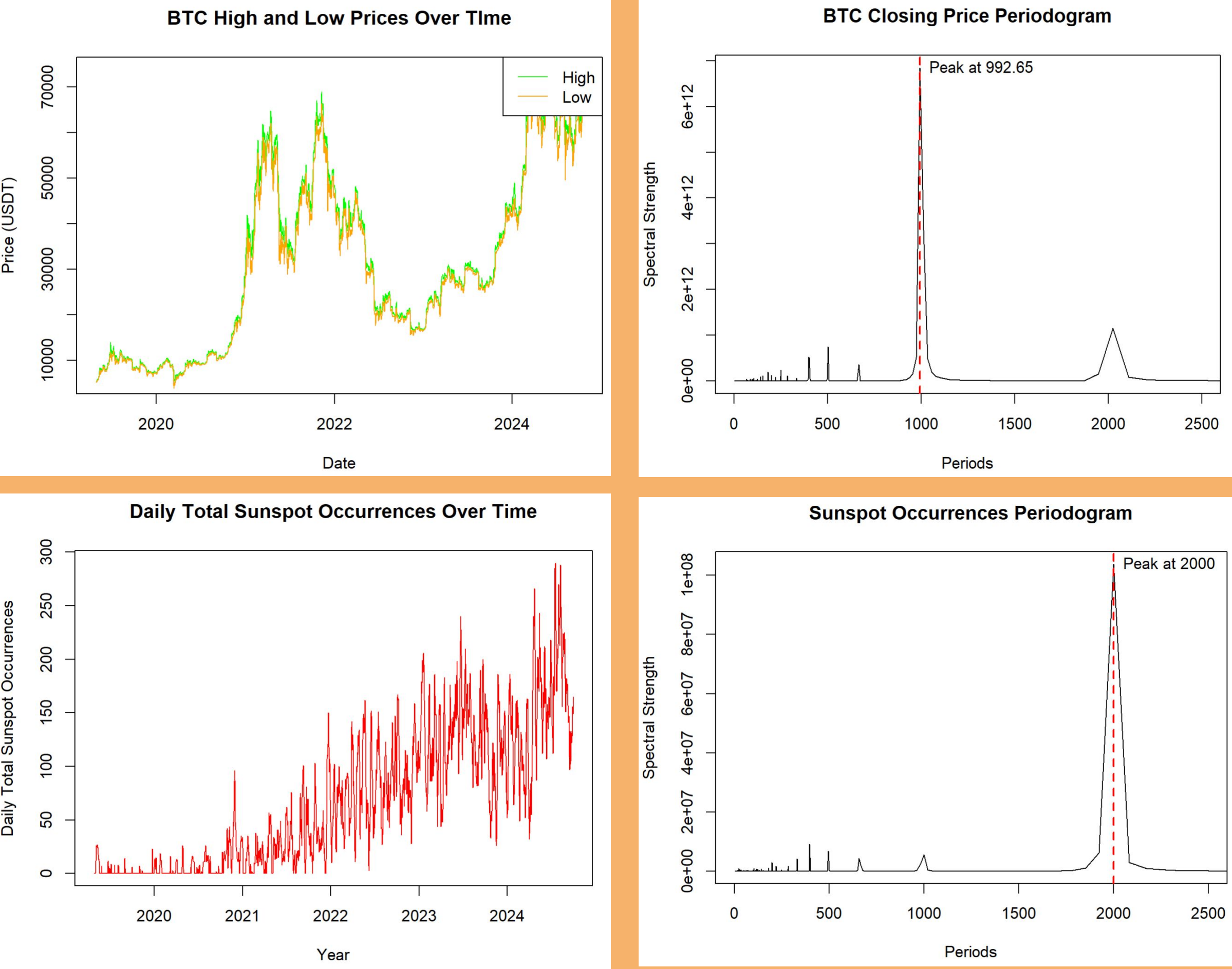


Figure 2: Raw data of Bitcoin (BTC) of high and low price over the studied data (top left). Periodogram with periods of Bitcoin close price with a peak at approximately 2.75 years, and some minor annual cycles (365 days; top right). Raw data of sunspots occurrences (bottom left), and the periodogram with periods of approximately 5.5 years (bottom right).

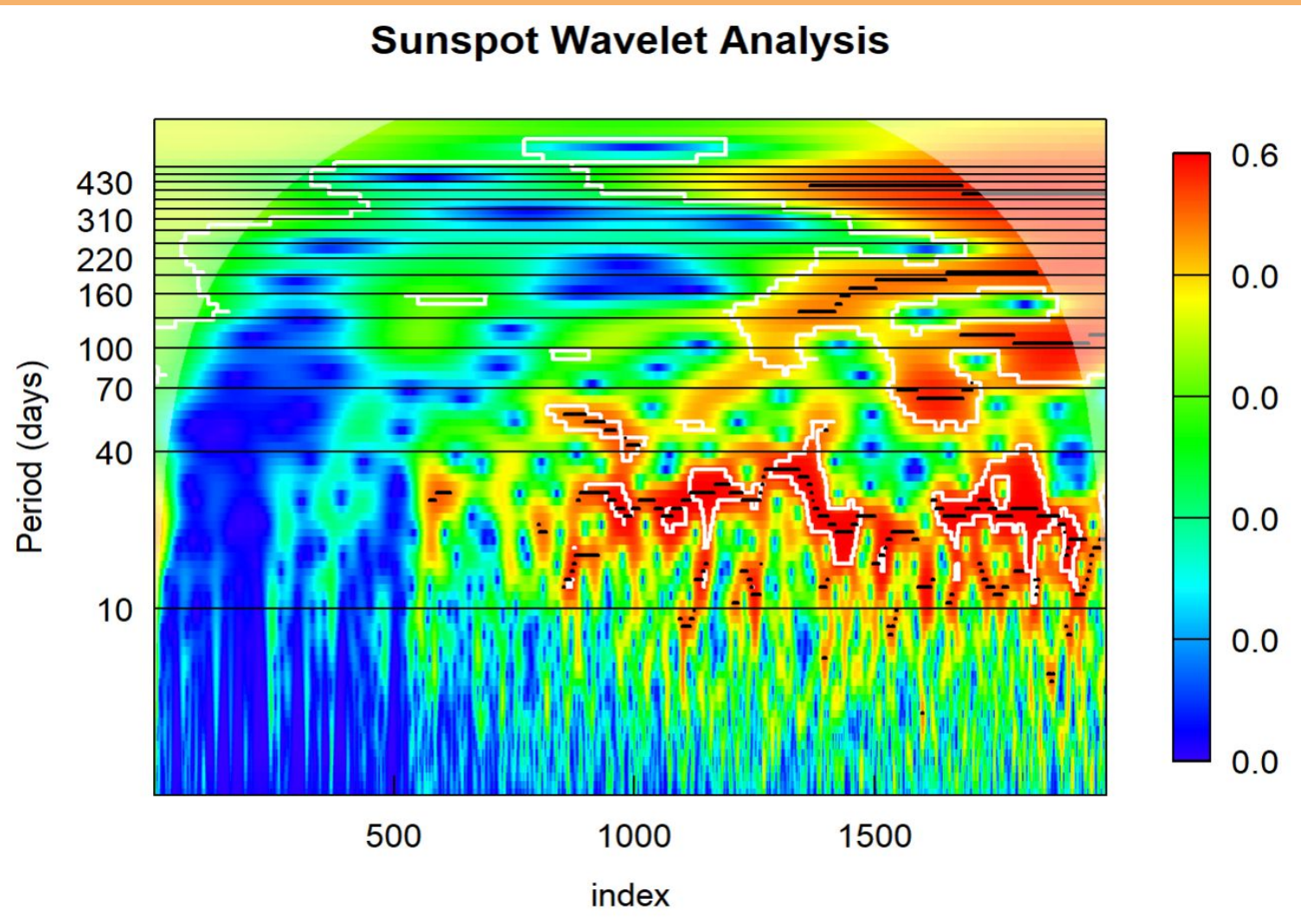


Figure 3: Wavelet analysis of sunspots number over the data range of analysis (x-axis).

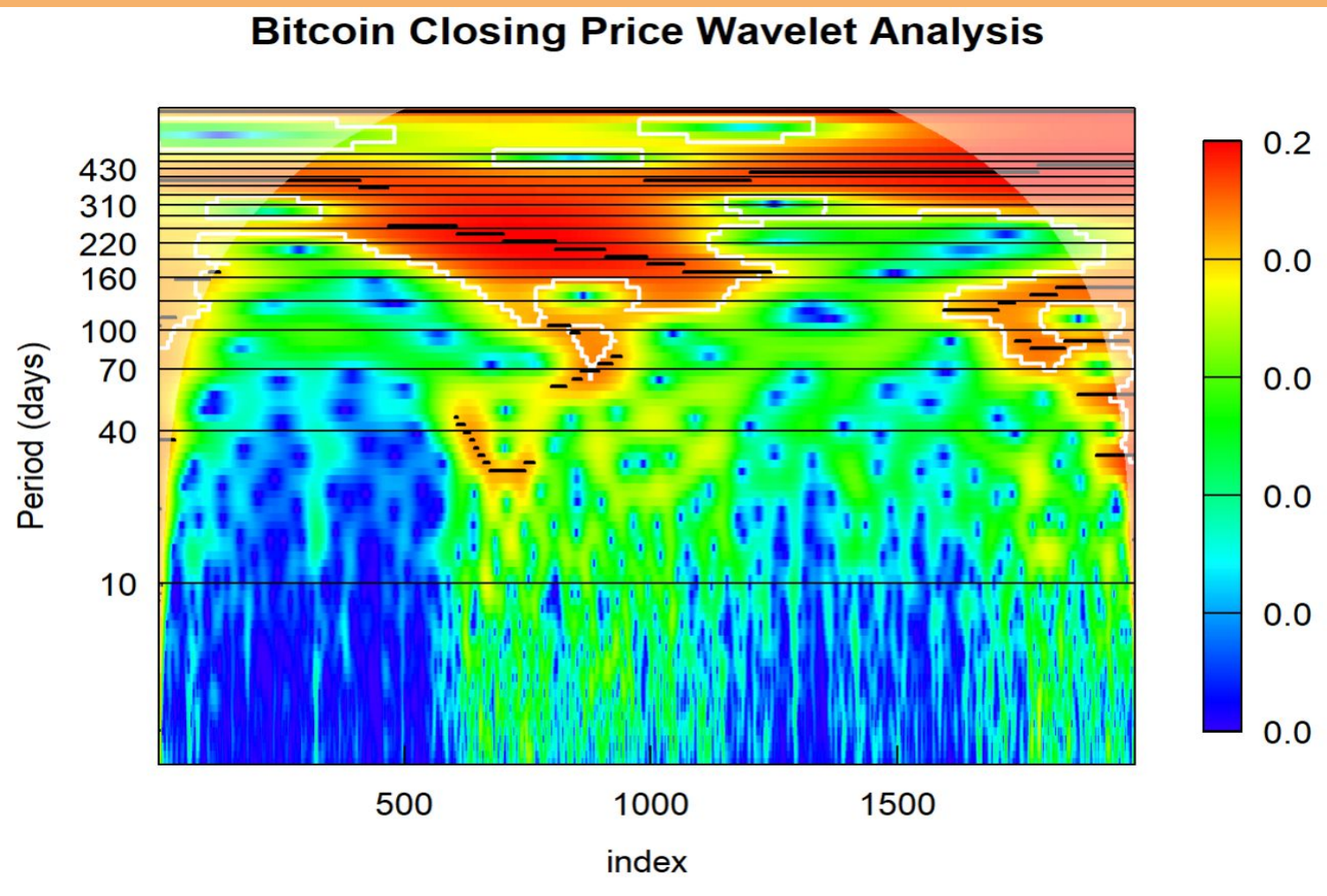


Figure 4: Wavelet analysis of bitcoin close price over the data range of analysis (x-axis).

Discussion/Conclusion:

- The noise inherent to human-driven systems like Bitcoin differs fundamentally from the periodicity of naturally occurring events like sunspot cycles.
- Human activity introduces complexity into datasets (e.g., financial markets), whereas natural phenomena follow predictable patterns. We explored the need for distinct analytical approaches to detect signals effectively in human-influenced versus natural datasets.
- **Summary of Key Findings:** Specialized methods are required to detect seasonality in human-driven data like Bitcoin, compared to natural data like sunspots, in order to best reveal patterns.
- **Future Directions:** Refine signal detection techniques for specialized data and explore additional datasets that involve both human-influenced and natural phenomena in order to confirm findings. Examine other types of data (with different types of noise) to find their most appropriate analysis method.

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References:

- Barrett T, Dowle M, Srinivasan A, Gorecki J, Chirico M, Hocking T (2024). `_data.table`: Extension of "data.frame". R package version 1.15.4, <<https://CRAN.R-project.org/package=data.table>>.
- Close B, Zurbenko I, Sun M (2020). `_kza`: Kolmogorov-Zurbenko Adaptive Filters. R package version 4.1.0.1, <<https://CRAN.R-project.org/package=kza>>.
- Dobrijevic, D. (2022, May 27). Sunspots: What are they, and why do they occur? Space.com. <https://www.space.com/sunspots-formation-discovery-observations>
- Garrett Grolemond, Hadley Wickham (2011). Dates and Times Made Easy with lubridate. Journal of Statistical Software, 40(3), 1-25. URL: <https://www.jstatsoft.org/v40/i03/>.
- Johnson-Groh, M. (2020, October 8). A New Look at Sunspots is Helping NASA Scientists Understand Major Flares and Life Around Other Stars - NASA. NASA.
- <https://www.nasa.gov/missions/hinode/a-new-look-at-sunspots-is-helping-nasa-scientists-understand-major-flares-and-life-around-other-stars/#:~:text=Sunspots%20are%20often%20precursors%20to>
- Marsellos, A., Tsakiri, K., & Tentomas, G. (2022). A GEOSPATIAL ANALYSIS OF THE SUNSPOTS EFFECT ON GROUNDWATER LEVELS IN LONG ISLAND, NY DURING 2002-2021. https://www.stonybrook.edu/commcms/geosciences/about/_LIG-Past-Conference-abstract-pdfs/2022-Abstracts/Marsellos.pdf
- Moritz S, Bartz-Beielstein T (2017). "imputeTS: Time Series Missing Value Imputation in R." *The R Journal*, 9*(1), 207-218. doi:10.32614/RJ-2017-009
- <<https://doi.org/10.32614/RJ-2017-009>>.
- Mourato, A., Faria, J., & Ventura, R. (2024). Automatic sunspot detection through semantic and instance segmentation approaches. Engineering Applications of Artificial Intelligence, 129, 107636. <https://doi.org/10.1016/j.engappai.2023.107636>
- Sunspots/Solar Cycle | NOAA / NWS Space Weather Prediction Center. (n.d.). [www.swpc.noaa.gov](https://www.swpc.noaa.gov/sunspots/solar-cycle).
- Tentomas, G., & Marsellos, A. (n.d.). SUNSPOT NUMBER EFFECTS ON EARTH'S CLIMATE CHANGE AND WATER SUSTAINABILITY FOR LONG ISLAND, NY. Retrieved August 26, 2024, from https://www.stonybrook.edu/commcms/geosciences/about/_LIG-Past-Conference-abstract-pdfs/2022-Abstracts/Tentomas.pdf
- US Department of Commerce, N. (n.d.). The Sun and Sunspots. [www.weather.gov](https://www.weather.gov/fsd/sunspots#:~:text=1%20sunspots%20are%20active%2C%20more).
- Wickham H, François R, Henry L, Müller K, Vaughan D (2023). `_dplyr`: A Grammar of Data Manipulation. R package version 1.1.4, <<https://CRAN.R-project.org/package=dplyr>>.
- Wickham H, Pedersen T, Seidel D (2023). `_scales`: Scale Functions for Visualization. R package version 1.3.0, <<https://CRAN.R-project.org/package=scales>>.
- Zhu, H., Chen, H., Zhu, W., & He, M. (2023). Predicting Solar cycle 25 using an optimized long short-term memory model based on sunspot area data. Advances in Space Research, 71(8), 3521-3531. <https://doi.org/10.1016/j.asr.2023.01.042>

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<https://docs.google.com/presentation/d/18PkwtWeGWPBtloto8QD88eidTHnI03MtZsg7A82Ds1c/edit>

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